THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 3: **Pair of Linear Equations in Two Variables** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** -2	Eliminate y
2	**(c)** 5	Eliminate x
3	**(c)** 2	All ratios equal
4	**(a)** 12	$\frac{k}{a_2} \neq \frac{b_1}{b_2}$ for a unique solution
5	**(c)** 9	Add the two equations
6	**(a)** 27	$(x+y)(x-y) = 9 \times 3$
7	**(b)** 4	Eliminate y
8	**(d)** 3	Eliminate x
9	**(c)** 3	All ratios equal
10	**(c)** 12	$\frac{k}{a_2} \neq \frac{b_1}{b_2}$ for a unique solution

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Write as $7x + 2y + 6 = 0$, $3x + 5y - 14 = 0$ (1); $\frac{x}{-58} = \frac{y}{116} = \frac{1}{29}$ (1); $x = -2$, $y = 4$ (1).	1+1+1
12	Unique solution when $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$ (1): $\frac{k}{6} \neq \frac{2}{1}$, i.e. $k \neq 12$ (1).	1+1
13	$\frac{2}{4} = \frac{k}{4} = \frac{6}{12}$ (1) gives $k = 2$ (1).	1+1
14	Write as $5x + 3y - 29 = 0$, $4x + 7y - 37 = 0$ (1); $\frac{x}{92} = \frac{y}{69} = \frac{1}{23}$ (1); $x = 4$, $y = 3$ (1).	1+1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Multiply (i) by 5 and (ii) by 2 and subtract: $(29)x = -58$ (2); $x = -2$ (1); put in (i): $y = 4$ (1); check in (ii) (1).	2+1+1+1
16	Let tens digit x, ones digit y: $x + y = 11$ (1); $(10y + x) - (10x + y) = 45$ so $y - x = 5$ (1); $x = 3$, $y = 8$, number 38 (1).	1+1+1
17	$\frac{k}{4} = \frac{1}{k}$ gives $k^2 = 4$ (1); $k = \pm 2$ (1); only $k = 2$ also satisfies $\frac{1}{k} = \frac{k-1}{k}$ (1).	1+1+1
	OR: Multiply (i) by 7 and (ii) by 3 and subtract: $(23)x = 92$ (2); $x = 4$ (1); put in (i): $y = 3$ (1); check in (ii) (1).	2+1+1+1
18	Let tens digit x, ones digit y: $x + y = 13$ (1); $(10y + x) - (10x + y) = 45$ so $y - x = 5$ (1); $x = 4$, $y = 9$, number 49 (1).	1+1+1
	OR: $\frac{k}{8} = \frac{2}{k}$ gives $k^2 = 16$ (1); $k = \pm 4$ (1); only $k = 4$ also satisfies $\frac{2}{k} = \frac{k-2}{k}$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Son x, father $10x$ (1); $10x + 6 = 4(x + 6)$ (2); $x = 3$ (1); son 3 years, father 30 years (1).	1+2+1+1
	OR: Let fraction $\frac{x}{y}$: $2(x + 1) = 1(y + 1)$ and $9(x + 2) = 5(y + 2)$ (2); simplify and solve (2); $x = 3$, $y = 7$, fraction $\frac{3}{7}$ (1).	2+2+1
20	Son x, father $6x$ (1); $6x + 10 = 5(x + 10)$ (2); $x = 40$ (1); son 40 years, father 240 years (1).	1+2+1+1
	OR: Let fraction $\frac{x}{y}$: $3(x + 3) = 2(y + 3)$ and $15(x + 6) = 11(y + 6)$ (2); simplify and solve (2); $x = 5$, $y = 9$, fraction $\frac{5}{9}$ (1).	2+2+1

Mistakes to avoid:

- Applying the ratio test with the wrong pairing, e.g. comparing $\frac{a_1}{b_2}$ instead of $\frac{a_1}{a_2}$.
- Saying a pair with $\frac{a_1}{a_2} = \frac{b_1}{b_2}$ always has no solution; check $\frac{c_1}{c_2}$ to separate coincident lines from parallel lines.
- Subtracting only some terms in elimination, or forgetting to multiply the right-hand side as well as the left.

- Reading the result $0 = 0$ as 'no solution'; it signals infinitely many solutions, while $0 = 5$ signals none.
- Not substituting back to find the second variable, or stating only x when the question asks for both.
- Writing the reversed number as $10x + y$ again instead of $10y + x$ in digit problems.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 3: **Pair of Linear Equations in Two Variables** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 8	Add the two equations
2	**(a)** 55	$(x+y)(x-y) = 11 \times 5$
3	**(c)** 5	Eliminate y
4	**(b)** 6	Eliminate x
5	**(c)** 4	All ratios equal
6	**(b)** 12	$\frac{k}{a_2} \neq \frac{b_1}{b_2}$ for a unique solution
7	**(d)** $\frac{60}{7}$	Add the two equations
8	**(c)** 52	$(x+y)(x-y) = 13 \times 4$
9	**(b)** 4	Eliminate y
10	**(c)** 2	Eliminate x

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Unique solution when $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$ (1): $\frac{k}{8} \neq \frac{3}{2}$, i.e. $k \neq 12$ (1).	1+1
12	$\frac{3}{6} = \frac{k}{6} = \frac{9}{18}$ (1) gives $k = 3$ (1).	1+1
13	Write as $6x + 5y - 60 = 0$, $7x + 4y - 59 = 0$ (1); $\frac{x}{-55} = \frac{y}{-66} = \frac{1}{-11}$ (1); $x = 5$, $y = 6$ (1).	1+1+1
14	Unique solution when $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$ (1): $\frac{k}{9} \neq \frac{4}{3}$, i.e. $k \neq 12$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Multiply (i) by 4 and (ii) by 5 and subtract: $(-11)x = -55$ (2); $x = 5$ (1); put in (i): $y = 6$ (1); check in (ii) (1).	2+1+1+1
16	Let tens digit x, ones digit y: $x + y = 9$ (1); $(10y + x) - (10x + y) = 45$ so $y - x = 5$ (1); $x = 2$, $y = 7$, number 27 (1).	1+1+1
17	$\frac{k}{12} = \frac{3}{k}$ gives $k^2 = 36$ (1); $k = \pm 6$ (1); only $k = 6$ also satisfies $\frac{3}{k} = \frac{k-3}{k}$ (1).	1+1+1
	OR: Multiply (i) by 3 and (ii) by 2 and subtract: $(17)x = 68$ (2); $x = 4$ (1); put in (i): $y = 7$ (1); check in (ii) (1).	2+1+1+1
18	Let tens digit x, ones digit y: $x + y = 7$ (1); $(10y + x) - (10x + y) = 45$ so $y - x = 5$ (1); $x = 1$, $y = 6$, number 16 (1).	1+1+1
	OR: $\frac{k}{16} = \frac{4}{k}$ gives $k^2 = 64$ (1); $k = \pm 8$ (1); only $k = 8$ also satisfies $\frac{4}{k} = \frac{k-4}{k}$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Son x, father $9x$ (1); $9x + 12 = 5(x + 12)$ (2); $x = 12$ (1); son 12 years, father 108 years (1).	1+2+1+1
	OR: Let fraction $\frac{x}{y}$: $13(x + 2) = 9(y + 2)$ and $15(x + 4) = 11(y + 4)$ (2); simplify and solve (2); $x = 7$, $y = 11$, fraction $\frac{7}{11}$ (1).	2+2+1
20	Fixed x, per km y: $x + 5y = 70$, $x + 12y = 140$ (2); subtract: $7y = 70$, $y = 10$ (1); $x = 20$ (1); fare for 20 km $= 20 + 10 \times 20 = 220$ (1).	2+1+1+1
	OR: Length x, breadth y: $(x - 3)(y + 2) = xy - 6$ gives $2x - 3y = 0$ (1.5); $(x + 2)(y + 2) = xy + 54$ gives $2x + 2y = 50$ (1.5); solving, $x = 15$, $y = 10$ (2).	1.5+1.5+2

Mistakes to avoid:

- Applying the ratio test with the wrong pairing, e.g. comparing $\frac{a_1}{b_2}$ instead of $\frac{a_1}{a_2}$.
- Saying a pair with $\frac{a_1}{a_2} = \frac{b_1}{b_2}$ always has no solution; check $\frac{c_1}{c_2}$ to separate coincident lines from parallel lines.
- Subtracting only some terms in elimination, or forgetting to multiply the right-hand side as well as the left.
- Reading the result $0 = 0$ as 'no solution'; it signals infinitely many solutions, while $0 = 5$ signals none.

- Not substituting back to find the second variable, or stating only x when the question asks for both.
- Writing the reversed number as $10x + y$ again instead of $10y + x$ in digit problems.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 3: **Pair of Linear Equations in Two Variables** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 5	All ratios equal
2	**(d)** 15	$\frac{k}{a_2} \neq \frac{b_1}{b_2}$ for a unique solution
3	**(c)** $\frac{101}{11}$	Add the two equations
4	**(a)** 105	$(x+y)(x-y) = 15 \times 7$
5	**(a)** 3	Eliminate y
6	**(d)** 9	Eliminate x
7	**(a)** 6	All ratios equal
8	**(b)** $\frac{5}{2}$	$\frac{k}{a_2} \neq \frac{b_1}{b_2}$ for a unique solution
9	**(c)** 10	Add the two equations
10	**(c)** 72	$(x+y)(x-y) = 12 \times 6$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\frac{5}{10} = \frac{k}{10} = \frac{15}{30}$ (1) gives $k = 5$ (1).	1+1
12	Write as $9x + 2y - 50 = 0$, $5x + 3y - 41 = 0$ (1); $\frac{x}{68} = \frac{y}{119} = \frac{1}{17}$ (1); $x = 4$, $y = 7$ (1).	1+1+1
13	Unique solution when $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$ (1): $\frac{k}{6} \neq \frac{5}{2}$, i.e. $k \neq 15$ (1).	1+1
14	$\frac{7}{14} = \frac{k}{14} = \frac{21}{42}$ (1) gives $k = 7$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Multiply (i) by 5 and (ii) by 3 and subtract: $(31)x = 93$ (2); $x = 3$ (1); put in (i): $y = 2$ (1); check in (ii) (1).	2+1+1+1
16	Let tens digit x, ones digit y: $x + y = 12$ (1); $(10y + x) - (10x + y) = 36$ so $y - x = 4$ (1); $x = 4$, $y = 8$, number 48 (1).	1+1+1
17	$\frac{k}{20} = \frac{5}{k}$ gives $k^2 = 100$ (1); $k = \pm 10$ (1); only $k = 10$ also satisfies $\frac{5}{k} = \frac{k-5}{k}$ (1).	1+1+1
	OR: Multiply (i) by 6 and (ii) by -7 and subtract: $(59)x = 472$ (2); $x = 8$ (1); put in (i): $y = 6$ (1); check in (ii) (1).	2+1+1+1
18	Let tens digit x, ones digit y: $x + y = 12$ (1); $(10y + x) - (10x + y) = 54$ so $y - x = 6$ (1); $x = 3$, $y = 9$, number 39 (1).	1+1+1
	OR: $\frac{k}{24} = \frac{6}{k}$ gives $k^2 = 144$ (1); $k = \pm 12$ (1); only $k = 12$ also satisfies $\frac{6}{k} = \frac{k-6}{k}$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Fixed x, per km y: $x + 10y = 130$, $x + 25y = 250$ (2); subtract: $15y = 120$, $y = 8$ (1); $x = 50$ (1); fare for 40 km $= 50 + 8 \times 40 = 370$ (1).	2+1+1+1
	OR: Length x, breadth y: $(x-6)(y+3) = xy - 42$ gives $3x - 6y = -24$ (1.5); $(x+4)(y+2) = xy + 120$ gives $2x + 4y = 112$ (1.5); solving, $x = 24$, $y = 16$ (2).	1.5+1.5+2
20	Fixed x, per km y: $x + 6y = 125$, $x + 14y = 245$ (2); subtract: $8y = 120$, $y = 15$ (1); $x = 35$ (1); fare for 22 km $= 35 + 15 \times 22 = 365$ (1).	2+1+1+1
	OR: Length x, breadth y: $(x-2)(y+3) = xy + 18$ gives $3x - 2y = 24$ (1.5); $(x+3)(y+1) = xy + 44$ gives $x + 3y = 41$ (1.5); solving, $x = 14$, $y = 9$ (2).	1.5+1.5+2

Mistakes to avoid:

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- Saying a pair with $\frac{a_1}{a_2} = \frac{b_1}{b_2}$ always has no solution; check $\frac{c_1}{c_2}$ to separate coincident lines from parallel lines.

- Subtracting only some terms in elimination, or forgetting to multiply the right-hand side as well as the left.
- Reading the result $0 = 0$ as 'no solution'; it signals infinitely many solutions, while $0 = 5$ signals none.
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